1. A tennis ball is thrown vertically downwards from a height of 0.65 m above the ground. It leaves the hand with an initial speed of 1.5 m s^{-1}. The ball rebounds and is caught when it is travelling upwards with a speed of 1.0 m s^{-1}.

Assume that air resistance is negligible.

(a) Calculate the speed of the ball just before it strikes the ground.

speed = ... m s^{-1} [2]

(b) The tennis ball is thrown downwards at t = 0. It hits the ground at time t_1 and is caught at time t_2.

On Fig. 1, sketch the velocity-time graph for the motion of the ball from the time it leaves the hand to when it returns.

Assume that the contact time between the ball and the ground is negligible. The initial velocity **X** and final velocity **Y** are marked on Fig. 1.

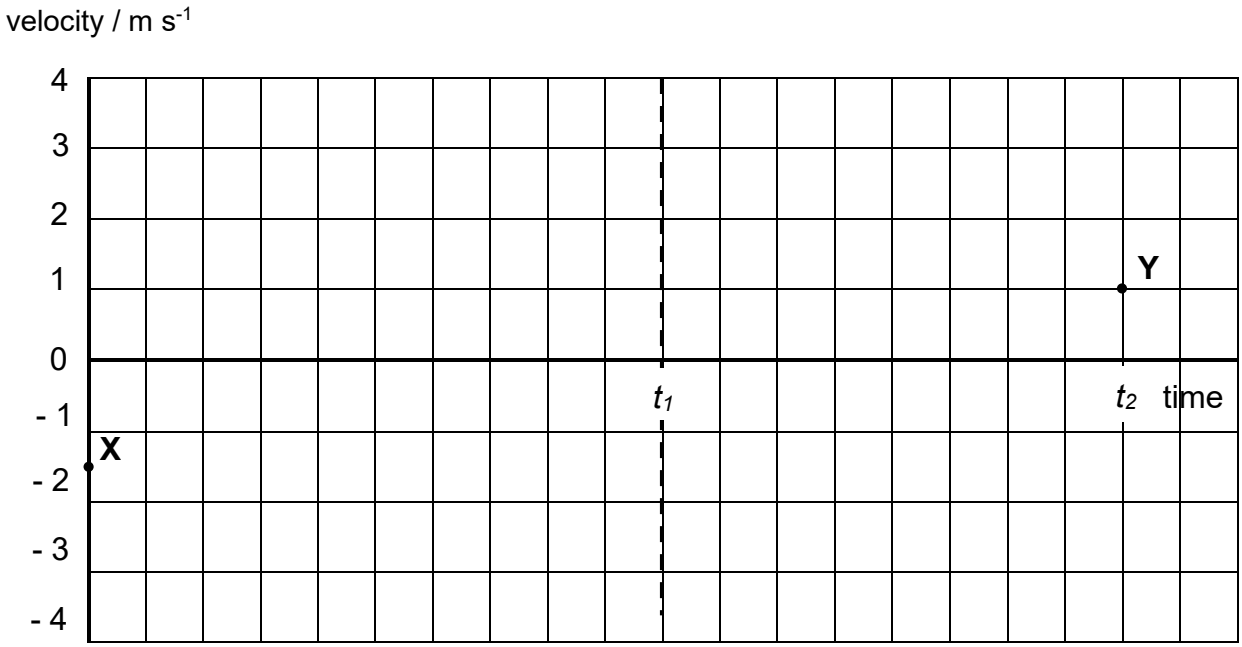

Fig. 1

[2]

(c) State and explain whether the bounce is elastic.

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[2]

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(d) On Fig. 1, sketch the velocity-time graph of the tennis ball if air resistance is not negligible. Label this graph **P**. [2]